

Climate Change and Nutritional Security: Which Crops Are Most Vulnerable?

A Climate-Nutrition Vulnerability Index for rice, wheat, maize and soybean

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Earth 616

 rice  wheat  maize  soybean

Problem and research gap

- Studies measure climate's effect on *yield* (Ray et al., 2015; Zhao et al., 2017) and CO₂'s effect on *nutrients* (Myers et al., 2014; Smith and Myers, 2018) separately.
- Their scores are on different scales, so crops cannot be ranked together.

Gap (from our literature review)

No study combines climatic **exposure**, crop **sensitivity** and **nutritional importance** in one index.

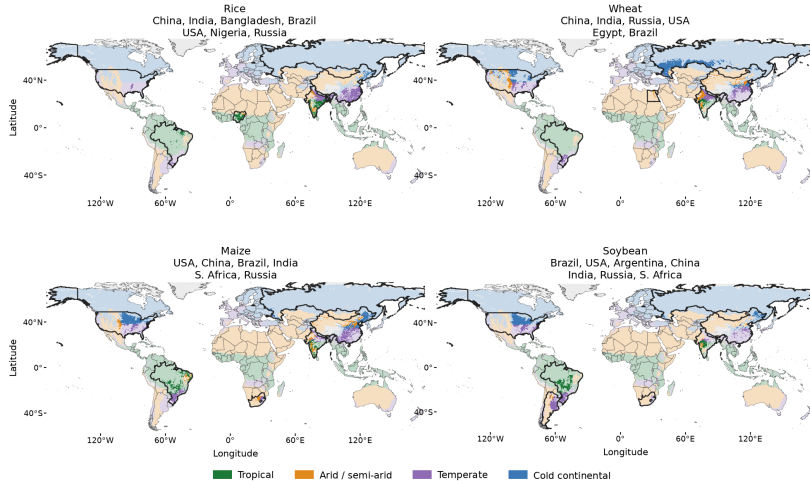
Our questions

1. Which crop is most vulnerable to observed temperature and precipitation change based on its nutritional importance?
2. If a crop's yield falls, what happens to a country's nutrient supply?

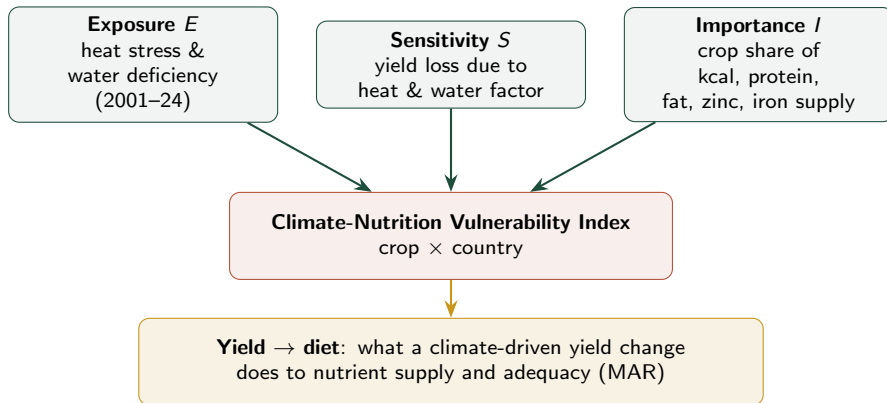
Countries analysed

● **Core:** USA, Brazil, India, China, Russia.

● **Others:** Top producers in different climate zones.



Framework: $\text{Vulnerability} = \text{Exposure} \times \text{Sensitivity} \times \text{Importance}$



IPCC-style framing with *Adaptive capacity* replaced by *Importance*.

Our formula for the Index

	Formula	Verdict
F1	$E \cdot S \cdot I$	Rejected: lets heat exposure multiply water sensitivity
F2	$(E S I)^{1/3}$	Rejected: same ranking as F1, only rescaled
F3	$(E + S + I)/3$	Rejected: high exposure can make up for a crop nobody eats
F4	$\frac{1}{2}(E^T S^T + E^P S^P) I$	Chosen: each stress meets its own sensitivity; zero if nobody eats the crop

The choice rests on these design reasons; calculations show that there is not much variance in the ordering of vulnerability score.

Exposure to temperature (heat)

Formula

$$E^T = \frac{L^T}{\max L^T}, \quad L^T = \text{mean}_{2001-24} \frac{\sum_m w_m \max(0, T_{\text{eff},m} - T_{\text{crit}})}{\sum_m w_m}$$

In words: the average number of °C by which flowering-season temperature is above the crop's heat limit; then divided by the largest value among all crop-country pairs, so everything sits on 0–1.

- m : month; w_m : share of month m inside the flowering window (45–70% of the season, Deryng et al. (2014))
- $T_{\text{eff}} = \frac{1}{2}(T_{\text{mean}} + T_{\text{max}})$: effective temperature
- T_{crit} : heat limit above which flowering is damaged

Crop	T_{crit}
Rice	35 °C
Wheat	25 °C
Maize	32 °C
Soybean	35 °C

Deryng 2014; rice: Yoshida 1981.

Exposure to precipitation (water)

Formula

$$E^P = \frac{L^P}{\max L^P}, \quad L^P = \text{mean}_{2001-24} \frac{\sum R \max(0, 1 - P/ET_c)}{\sum A}$$

In words: the share of the crop's water need that rain does not supply, counted on rainfed land; then divided by the largest value among all pairs (0–1).

- P : rainfall in the growing season
- ET_c : the crop's water need. (Allen et al., 1998)
- $1 - P/ET_c$: the shortfall (0 if rain is enough)
- R : rainfed hectares, A : all hectares. Irrigated land counts as no shortfall

Why this form: it is FAO's own idea (a crop's yield loss follows its water deficit (Steduto et al., 2012)) computed from monthly rainfall and temperature, so it also runs on future climate data.

Sensitivity to temperature (heat)

Formula

$$S^T = \frac{s^T}{\max s^T}, \quad s^T = \frac{1}{T_{lim} - T_{crit}}$$

In words: fraction of yield lost per °C above the crop's heat limit, divided by the largest of the four crops (0–1).

- T_{crit} : heat limit, where damage starts (same as in exposure)
- T_{lim} : total loss; linear in between (Deryng et al., 2014)

Why it fits: exposure = degrees above the limit;
sensitivity = yield lost per degree above the limit; the
product is the fraction of yield at risk.

Crop	T_{crit}	T_{lim}	loss per °C (%)	S^T
Rice	35	–	7.0	0.35
Wheat	25	35	10.0	0.50
Maize	32	45	7.7	0.38
Soybean	35	40	20.0	1.00

Sensitivity to precipitation (water)

Formula

$$S^P = \frac{K_y}{\max K_y}$$

In words: FAO's water factor K_y : yield lost per unit of water shortfall, divided by the largest crop value (0–1).

- FAO (Steduto et al., 2012): yield loss = $K_y \times$ water deficit; $K_y = 1.25$: a 10% shortfall costs about 12.5% of yield
- Same quantity as exposure (the unmet share of water need), so the two multiply cleanly

Crop	K_y	S^P
Rice	1.25	1.00
Wheat	1.10	0.88
Maize	1.25	1.00
Soybean	0.85	0.68

Importance: how much do people depend on the crop?

Importance of a crop in a country

$$I = \frac{1}{5} \left(\text{share of kcal} + \text{protein} + \text{fat} + \text{zinc} + \text{iron} \right)$$

share = amount supplied by the crop / total supply (2021–23)

Data: FAO Food Balance Sheets; zinc and iron = food quantity × USDA content. Example: rice gives Bangladesh 63% of its calories and 55% of its zinc.

Why an average?

- Nothing ranks calories above zinc, so equal weights are neutral
- An average lets a crop count where it is strong (soybean: 12.5% of world fat, 1.3% of zinc)
- A geometric mean would punish it for its weakest nutrient.

From yield to diet

1. History: pass-through

$$\Delta\% \text{ food supply}_t = a + \beta \Delta\% \text{ yield}_t + \varepsilon_t$$

- Take yield and food supply as % above or below their long-term trend
- $\beta = 1$: food supply falls as much as yield
- $\beta \approx 0$: trade or stocks cover the gap
- Shows a link, not a cause

2. What-if: yield falls 10%

$$f = |\Delta Y| \min(1, \text{SSR}), \quad S'_n = S_n - f C_n$$

- Suppose yield falls 10%
- $\text{SSR} = \text{own production} \div \text{domestic supply}$
- The crop loses a share f of its calories, protein, fat, Zn and Fe (C_n = what the crop supplies)
- MAR is recomputed (R_n = requirement); 1 = all needs met
- No market response: prices, trade and stocks stay fixed; other foods unchanged

Scenario: from a yield loss to nutritional adequacy

Assumption: we do not model market responses. We assume no change in prices, trade, imports, exports or stocks after a yield shock, and other foods remain unchanged.

1. Yield shock → food-supply loss

$$f = |\Delta Y| \min(1, \text{SSR})$$
$$\text{SSR} = \frac{\text{domestic production}}{\text{domestic food supply}}$$

- ΔY = % change in yield; SSR = self-sufficiency ratio (baseline data); $\min(1, \text{SSR})$ caps the factor at 1
- **Example:** $\Delta Y = -10\%$, $\text{SSR} = 0.8$:
 $f = 0.10 \times 0.8 = 0.08$, i.e.

8% food-supply loss

Yield loss → food-supply loss → nutrient loss →
lower NAR → lower MAR

2. Loss → nutrients

$$S'_n = S_n - fC_n$$

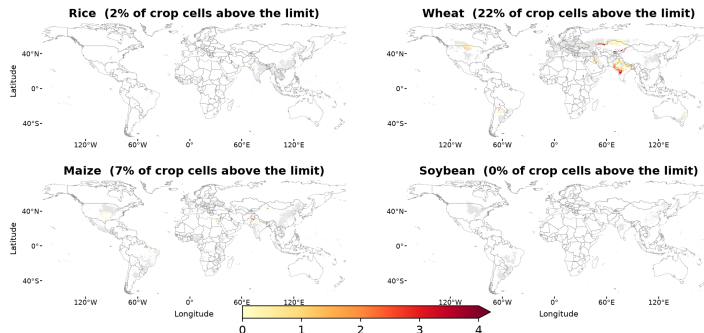
S_n = national supply of nutrient n ; C_n = supplied by the crop; S'_n = after the shock. The same fraction f of the crop's kcal, protein, fat, Zn and Fe is removed.

3. Recalculate adequacy

$$\text{NAR}_n = \min\left(1, \frac{S'_n}{R_n}\right), \quad \text{MAR} = \frac{1}{5} \sum_n \text{NAR}_n$$

- R_n = requirement; $n \in \{\text{kcal, protein, fat, Zn, Fe}\}$
- NAR_n capped at 1: a surplus of one nutrient cannot offset a deficiency of another
- $\text{MAR} = 1$: all five requirements met

Heat stress where each crop grows



Heat stress 2001-24: degrees (°C) by which flowering-season temperature exceeds the crop's heat limit (grey = crop grows, no excess)

What we measure:

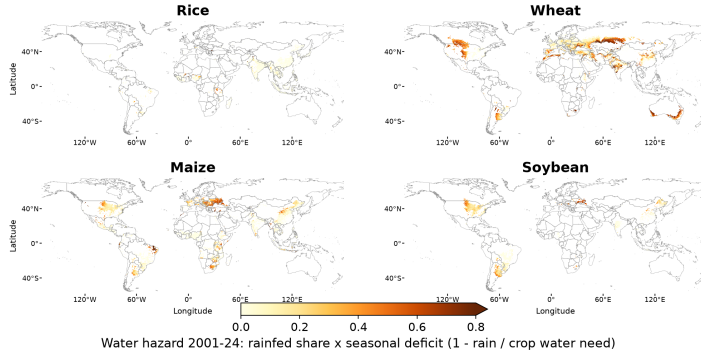
Flowering-season temperature excess above crop heat limit T_{crit} (2001–24 mean). Heat damage concentrates during flowering.

Crop	Limit	Cells above
Rice	35 °C	2%
Wheat	25 °C	22%
Maize	32 °C	7%
Soybean	35 °C	0%

Main finding:

Wheat is the most heat-exposed globally, largely because its threshold (25 °C) is much lower than other crops.

Water stress where each crop grows



What we measure:

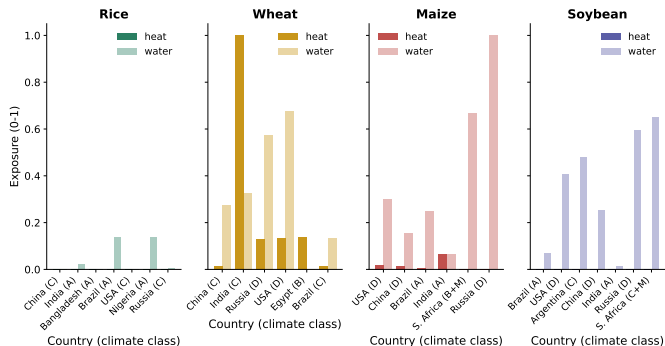
water stress = rainfed share $\times \max(0, 1 - P/ET_c)$

- $P \geq ET_c$: zero shortfall; irrigated land: zero stress.

Main finding & why it matters:

- Stress concentrates in dry, rainfed regions; irrigated rice/wheat show little.
- Vulnerability depends not just on rainfall, but on crop reliance on rain.

Exposure by country



Heat (E^T):

Peaks for wheat in India (1.00) where flowering temperatures exceed 25 °C.

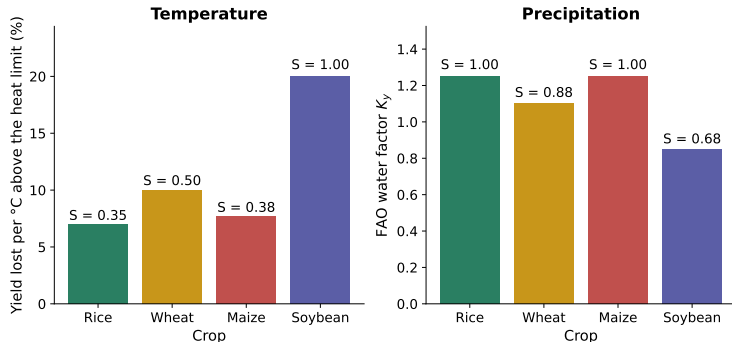
Water (E^P):

Highest for maize in Russia/South Africa and wheat in USA/Russia (dry rainfed zones).

Rice:

Low on both: mostly irrigated and monthly means stay below its heat threshold.

Sensitivity by crop



Heat (S^T):

Soybean is the most sensitive per °C above threshold, then wheat, maize, rice.

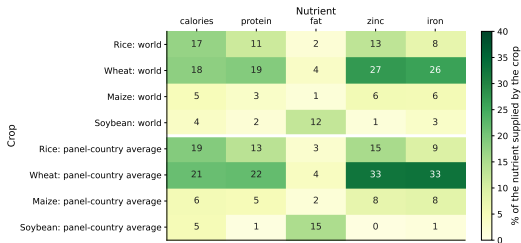
Water (S^P):

Rice and maize exhibit the highest FAO water factors ($K_y = 1.25$).

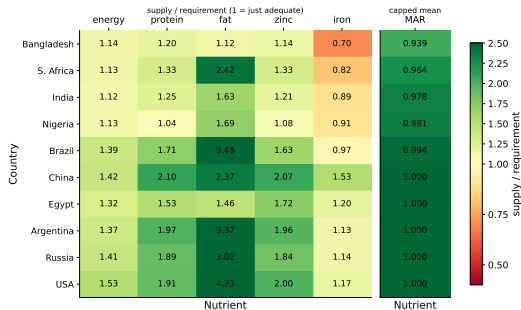
One score per crop:

Same literature values applied globally; sensitivity varies by crop, not country.

Importance and adequacy



% of each nutrient supplied by the crop



supply ÷ requirement (1 = just enough)

Result: wheat and rice supply most protein, zinc and iron; soybean mainly fat (oil); maize little (mostly feed). **Iron is limiting:** supply ÷ requirement 0.70 (Bangladesh), 0.82 (South Africa), 0.89 (India); other nutrients are met. **Why it matters:** a yield loss in a crop people depend on, where a nutrient is already short, hurts nutrition most.

Country-by-crop vulnerability index (V_{F4})

Country	Rice	Wheat	Maize	Soybean
USA	< 0.0001	0.0747	0.0037	0.0197
Brazil	0.0039	0.0103	0.0103	0.0025
India	0.0023	0.1511	0.0015	0.0002
China	0.0001	0.0240	0.0016	0.0097
Russia	< 0.0001	0.1105	0.0021	0.0032
Argentina	–	–	–	0.0052
Bangladesh	0.0002	–	–	–
Egypt	–	0.0209	–	–
Nigeria	0.0046	–	–	–
South Africa	–	–	0.0895	0.0122

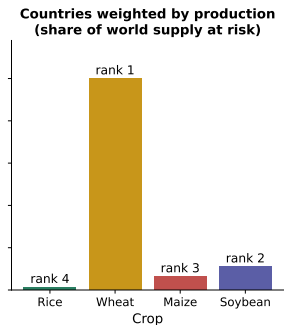
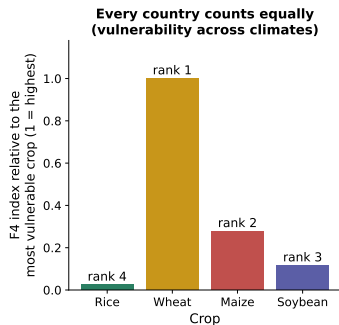
Why values are on a small decimal scale:

$V_{F4} = \frac{1}{2}(E^T S^T + E^P S^P) /$ multiplies three fractional shares (≤ 1), yielding values from 10^{-5} to 0.15.

Key patterns:

- **Wheat peaks:** in India (0.1511) and Russia (0.1105) from high dietary shares and climate exposure.
- **Maize peaks:** in South Africa (0.0895) where it is a dietary staple on rainfed land.
- **Rice stays low:** (< 0.005) across all panel countries due to low observed heat exposure.

Result: the vulnerability index



Equal weights: Wheat > Maize > Soybean > Rice

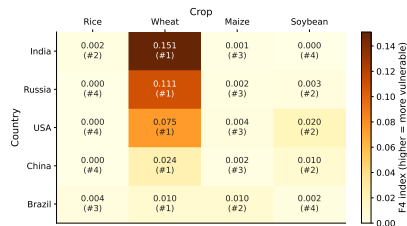
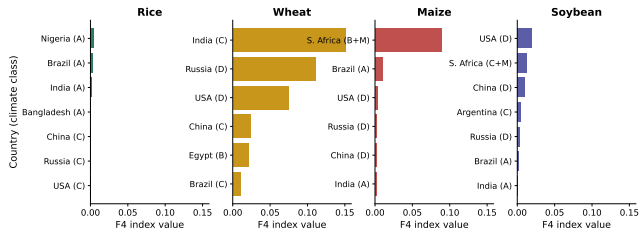
Production-weighted: Wheat > Soybean > Maize > Rice

Bar = index relative to the most vulnerable crop (= 1).

Wheat (#1 in both): main source of protein, zinc, iron; exposed to heat in India and dry spells in USA/Russia.

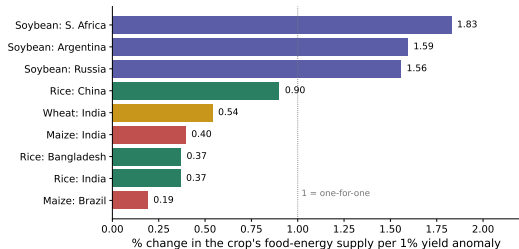
Rice (lowest in both): mostly irrigated; monthly means stay below its 35 °C limit.

Where does the vulnerability actually sit?

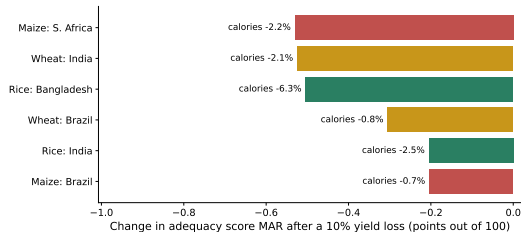


Left figure: each bar is 1 country's index for a particular crop. Right figure: crops compared within the 5 countries that grow all 4 crops (#1 here means the most vulnerable crop there and so on...); wheat ranks #1 in all 5 out of 5 countries. Highest: wheat in India, Russia, USA; maize in South Africa.

Yield to diet: results



pass-through β (clear cases only)



fall in MAR after a 10% yield loss

Left figure: a bad harvest clearly lowers food supply in 9 of 26 cases, mainly rice in Asia and small soybean crops. Right figure: maize in South Africa and wheat in India lose most adequacy (they remove iron where it is short); rice in Bangladesh loses most calories (6.3%). Exports can soften the loss (wheat in India: $-0.52 \rightarrow -0.28$).

Observations

1. **Wheat is the most vulnerable crop:** Wheat > Maize > Soybean > Rice (equal weights).
2. **Why wheat:** importance and exposure. Maize and soybean are the most heat-sensitive (-11.7% and -6.2% yield per °C; wheat -2.7%) but little is eaten directly.
3. **Most vulnerable pairs:** wheat in India, Russia, USA; maize in South Africa.
4. **Iron** is the nutrient closest to running short.
5. **Rice scores lowest because of the measure, not proof of safety:** mostly irrigated, and monthly means stay below its 35 °C limit though short heat spikes can still harm flowering. Where rice is the staple (Bangladesh: 63% of calories) an underestimate would matter.

Data used

Dataset	What it gives	Index term
WorldClim 2.1	monthly Tmax, Tmin, rainfall	Exposure
SPAM 2010	where each crop grows; irrigated vs rain-fed	Exposure
Sacks crop calendar	planting and harvest dates	Exposure
FAOSTAT crops	production, area, yield	Countries, Sensitivity
FAO Food Balance Sheets	food supply of calories, protein, fat, each food	Importance
USDA FoodData Central	zinc and iron per food	Importance
FAO energy requirement	average requirement per person	Adequacy

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